ANTOCCI ENGINEERING · DWG U1-16 · R9

RAMP SPEC: LOADING DOCK

A 2 kg crate slides on a 30° ramp. g = 9.8 m/s². Weight = 19.6 N.

CALC 1 · DOWN-RAMP PART

$mg \sin \theta = 19.6 \times \sin 30° =$ ____ N

CALC 2 · INTO-RAMP PART

$mg \cos \theta = 19.6 \times \cos 30° =$ ____ N

DRAWING · RAMP FBD: weight straight down, normal ⊥ surface, friction up the ramp. Dashed component guides.

CALC 3 · NO FRICTION

$a = (mg \sin \theta) \div m =$ ____ m/s² down the ramp

CALC 4 · WITH 4 N FRICTION

$F_{net} =$ ____ − 4 = ____ · a = ____

ENGINEER'S NOTE

Is the normal force equal to the weight on this ramp? Explain for the dock crew.

R9 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: net force from the FBD

16 C

DIRECTIONS · Solve each station. Copy the letter into the lock.

STATION 1

Downhill part 6 N, friction 2 N uphill, m = 2 kg. a?

(A) 4 m/s² (T) 2 m/s²

STATION 2

Downhill part 8 N, friction 2 N, m = 3 kg. a?

(I) 2 m/s² (O) 3 m/s²

STATION 3

Down-ramp part of weight is...

(L) mg sin θ (N) mg cos θ

STATION 4

2 kg on 30°: mg sin θ (g = 9.8)?

(T) 9.8 N (P) 17.0 N

STATION 5

Frictionless 30° ramp. a = g sin θ?

(E) 4.9 m/s² (U) 9.8 m/s²

STATION 6

Normal force on a ramp equals...

(D) mg cos θ (S) mg

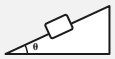
LOCK 1 2 3 4 5 6

Hint: how a ramp's surface sits.

R9 1 2 3

Say it, then write it: forces on a ramp

Language goal: I can **explain** which way it accelerates.



incline
plano inclinado · plano inclinado
a ramp



perpendicular
perpendicular · perpendicular
at a right angle



parallel
paralelo · paralelo
along the same line



net force
fuerza neta · força resultante
all forces added up

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: perpendicular · parallel · down · net · acceleration

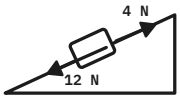
On a ramp, the normal force is ____ to the surface. Part of the weight is ____ to the ramp and points ____ the ramp. Friction points up the ramp. The ____ force decides the direction of the ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The net force points ____, so the box accelerates ____ .
The normal force is not straight up because ____ .

3 · SOLVE WITH THE STEPS



A 2 kg box on a ramp: 12 N down the ramp, 4 N friction up the ramp.

STEP 1 · NET FORCE
____ − ____ = ____ N down

STEP 2 · DIVIDE
a = ____ ÷ ____ = ____ m/s²

STEP 3 · SAY IT
"The box speeds up at ____ m/s², down the ramp."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ The box speeds up because the net force ____ .
☐ I split the weight into parts because ____ .
☐ First I find the ____ force, then the ____ .

I said it ☐ · I wrote it ☐

R9 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every ramp

16 M

DIRECTIONS · Read, draw the FBD, solve along the ramp, then write.

PROBLEM 1 · A 2 kg block has a 7 N downhill weight component and 3 N friction uphill. Find its acceleration.

1 READ

m ____ · downhill ____ · uphill ____

2 DRAW

ramp FBD

3 SOLVE

F_net = 7 − 3 · a = F_net ÷ m

4 WRITE

The block accelerates ____ m/s² ____ because ____ .

PROBLEM 2 · A 2 kg crate is on a frictionless 30° ramp (g = 9.8). Find mg sin θ, then a.

1 READ

m ____ · θ ____ · g ____

2 DRAW

ramp FBD with axes

3 SOLVE

mg sin θ → a = F ÷ m

4 WRITE

The crate speeds up at ____ m/s². The normal force is ____ the weight because ____ .

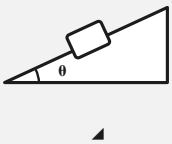
Hardest box: □ Read □ Draw □ Solve □ Write

R9 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: ramps

Language goal: I can **describe** forces on a slope.



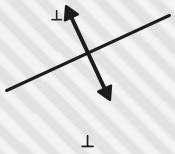
incline

plano inclinado · plano inclinado

a tilted surface

my language: _____

use it: _____



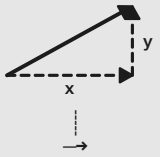
perpendicular

perpendicular · perpendicular

at a right angle

my language: _____

use it: _____



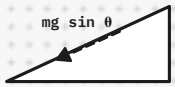
component

componente · componente

one part of a force

my language: _____

use it: _____



mg sin θ

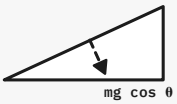
down the ramp

cuesta abajo · rampa abaixo

pulls the block downhill

my language: _____

use it: _____



mg cos θ

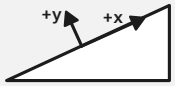
into the ramp

hacia la rampa · contra a rampa

presses into the surface

my language: _____

use it: _____



+x ↘
+y ↗

axis

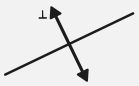
eje · eixo

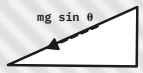
the direction we measure

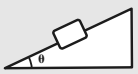
my language: _____

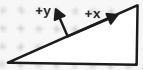
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

R9 1 2 3

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UNIT 1 · DAY 16 · R9 · LOOP CHAIN

I have... who has? Ramps and net force

16 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, and keep going.

★ START — **A**

WHO HAS…

Downhill part of weight on a ramp?

I have **4.9 m/s²** — **B**

WHO HAS…

7 N downhill, 3 N friction, 2 kg: a?

I have **mg cos θ** — **C**

WHO HAS…

2 kg, 30°, g = 9.8: mg sin θ?

I have **m a_x** — **D**

WHO HAS…

Which way does weight always point?

I have **mg sin θ** — **E**

WHO HAS…

Into-ramp part of weight?

I have **straight down** — **F**

WHO HAS…

Does normal force always equal weight?

I have **9.8 N** — **G**

WHO HAS…

That crate, frictionless: a?

I have **2 m/s²** — **H**

WHO HAS…

Each axis gets its own equation: $\Sigma F_x = ?$

I have **no — only if level** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

R9 1 2 3

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Comic: gravity still pulls straight down

16 CS

YOUR READER

A younger student who thinks gravity pulls a box "down the ramp."

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ ORANGE — The weight arrow pointing straight down
- ☐ ◌ ORANGE — Dashed component arrows along and into the ramp
- ☐ ◌ ORANGE — The normal force arrow, perpendicular to the ramp
- ☐ ◯ PENCIL — A right-angle mark between the components

1 · THE RAMP *A box sitting on a ramp.*

CAPTION _____

2 · STRAIGHT DOWN *Show which way gravity really pulls.*

CAPTION _____

3 · SPLIT IT *Break the weight into "along" and "into" the ramp.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **weight · component · normal force · perpendicular**

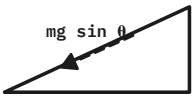
R9 1 2 3

Doodle Notes · force component

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Force component · componente de una fuerza (ES) · componente de uma força (PT)



TODAY'S CASE

A crate rests on a ramp. Choose axes parallel and perpendicular to its surface.

01 · SKETCH AND LABEL

Draw the weight straight down in orange, then add both ramp axes in pencil.

02 · CHANGE THE PICTURE

On a separate sketch resolve weight along the two axes with dashed projections. Explain why those projections do not add two new forces.

Annotate your sketches. Circle the mark that best shows force component. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

Tilting the axes changes ___ but not …

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Frayer Model · force component

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Force component · componente de una fuerza (ES) · componente de uma força (PT)

A · Part of weight acts parallel to the ramp axis.

B · The two weight components are new forces in addition to weight.

01 · DEFINITION

What does force component mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

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